Part V Fourier Transform

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5-1 Fourier Series

$$f(t) = a_0 + \sum_{n=1}^{\infty}[a_n \cos n\omega_0 t + b_n \sin n\omega_0 t]$$

where a_0 is the dc term, and the latter one is the ac components

$$a_0 = \frac{1}{T}\int_0^T f(t)\mathrm{d}t$$
$$a_n = \frac{2}{T}\int_0^T f(t)\cos n\omega_0 t\mathrm{d}t$$
$$b_n = \frac{2}{T}\int_0^T f(t)\sin n\omega_0 t\mathrm{d}t$$

Magnitude-Phase Form

$$f(t) = a_0 + \sum_{n=1}^{\infty}(A_n \cos(n\omega_0 t + \phi_n))$$

- A_n: $\sqrt{a_n^2 + b_n^2}$
- ϕ_n: $-\tan^{-1}(b_n/a_n)$

Effects of Symmetry on Fourier coefficients

Symmetry	a_0	a_n	b_n
Even	/	/	0
Odd	0	0	/
Half-wave	0	$a_{2n}=0$	$b_{2n}=0$
Quarter-wave	0	0	$b_{2n}=0$

5-2 Circuits and Fourier Series

Fourier series approaches can be used where a **repetitive non-sinusoidal** forcing function occurs and the Natural Response is not important

1. Decide on the upper limit of n: N_{max}
2. Find the FS of the **forcing function** for $n \in [0, N_{max}]$
3. Find the **transfer function** of the circuit
4. Solve for each component of the Forced Response using phasor method
5. Combine the response to each harmonic frequency to give the overall response

5-3 Average Power

$$v(t) = V_{dc} + \sum_{n=1}^{\infty} \cos(n\omega_0 t - \theta_n) \qquad i(t) = I_{dc} + \sum_{n=1}^{\infty} I_m \cos(m\omega_0 t - \phi_m)$$

then substitute them into

$$\begin{aligned} P &= \int_0^T v(t)i(t)\mathrm{d}t \\ &= V_{dc}I_{dc} + \frac{1}{2}\sum_{n=1}^{\infty} V_n I_n \cos(\theta_n - \phi_n) \end{aligned}$$

5-4 Exponential form of the Fourier Series

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}$$

the function could also be written as the pattern above, where

$$c_n = \frac{1}{T}\int_0^T f(t)e^{-jn\omega_0 t}\mathrm{d}t$$

5-5 Fourier Transform

$$F(j\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}$$

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(j\omega)e^{j\omega t}\mathrm{d}\omega$$

Transform Pairs and Properties Table

Aperiodic Signal	Fourier Signal	Signal	Fourier Transform
$ax(t) + by(t)$	$aX(j\omega) + bY(j\omega)$	$\sum_{k=-\infty}^{\infty} a_k e^{j\omega_0 t}$	$2\pi \sum_{k=-\infty}^{\infty} a_k \delta(\omega - k\omega_0)$
$x(t - t_0)$	$e^{-j\omega t_0} X(j\omega)$	$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$
$e^{j\omega_0 t} x(t)$	$X(j(\omega - \omega_0))$	$\cos \omega_0 t$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
$x^*(t)$	$X^*(-j\omega)$	$\sin \omega_0 t$	$\frac{\pi}{j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$
$x(-t)$	$X(-j\omega)$	$x(t) = 1$	$2\pi\delta(\omega)$
$x(at)$	$\frac{1}{\lvert a \rvert} X(\frac{j\omega}{a})$	$x(t) = \begin{cases} 1 & \lvert t \rvert < T_1 \\ 0 & \lvert t \rvert > T_1 \end{cases}$	$\frac{2 \sin \omega T_1}{\omega}$
$x(t) * y(t)$	$X(j\omega)Y(j\omega)$	$\frac{\sin Wt}{\pi t}$	$X(j\omega) = \begin{cases} 1 & \lvert \omega \rvert < W \\ 0 & \lvert \omega \rvert > W \end{cases}$
$x(t)y(t)$	$\frac{1}{2\pi} X(j\omega) * Y(j\omega)$	$\delta(t)$	1
$\frac{\mathrm{d}}{\mathrm{d}t} x(t)$	$j\omega X(j\omega)$	$u(t)$	$\frac{1}{j\omega} + \pi\delta(\omega)$
$\int_{-\infty}^{t} x(t)\mathrm{d}t$	$\frac{1}{j\omega} X(j\omega) + \pi X(0)\delta(\omega)$	$\delta(t - t_0)$	$e^{j\omega t_0}$
$tx(t)$	$j\frac{\mathrm{d}}{\mathrm{d}\omega} X(j\omega)$	$e^{-at}u(t)$	$\frac{1}{a + j\omega}$
$X(jt)$	$2\pi x(-\omega)$	$te^{-at}u(t)$	$\frac{1}{(a + j\omega)^2}$